

Synthetic Dental X-Ray Generation and Segmentation Analysis

18 February 2026
Computer Vision's course project

Authors:

Davide Patriarca [1910558] & **Ch**ristian Pizzuti [1872078]

Context & Challenges

Privacy & Ethics

Medical analysis faces strict challenges due to privacy restrictions and ethical concerns

OPT Scarcity

Obtaining diverse Orthopantomographic X-Ray (OPT) datasets is difficult due to patient data sensitivity

The Challenge

Real data is often insufficient or too sensitive for broad research use

Project Goal

GAN Synthesis

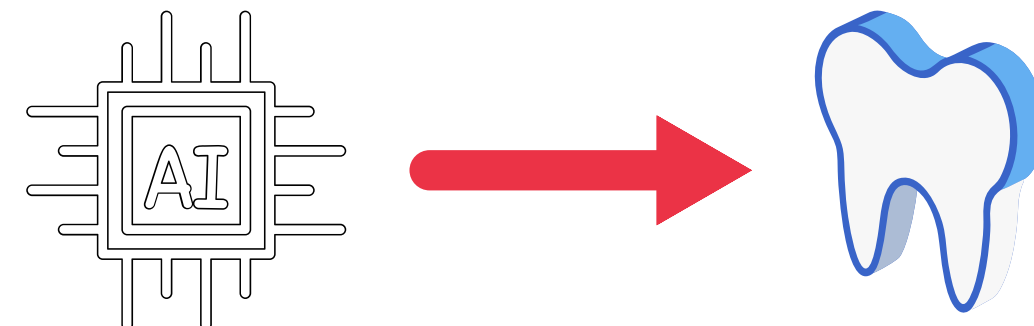
Train a GAN to synthesize realistic OPT images from segmentation masks

Validation Strategy

Evaluate effectiveness by training a CNN-based segmentation model on public dental datasets

Performance Assessment

Test on real vs. synthetic images to measure performance gaps, biases and usability



Hardware & settings

We use an Nvidia GeForce RTX 2060 as GPU



We decided not to rely on the standard Google Colab hosted runtime, but we utilized the Google Colab interface strictly as a frontend for code execution and visualization, while connecting it to a local runtime backed by our own hardware

Dataset

Source Dataset (Public Dental X-Ray)

Modality: OPT (Orthopantomographic X-Ray)

Size: 2041 x 1024 px (High Res)

Samples: 598 images

Three different type of files:

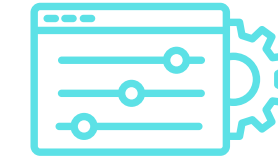
Images (.png): High-resolution panoramic dental X-rays

Vector Annotations (.json): Contains precise polygon coordinates (x,y) defining the boundary of each tooth

Masks (.png): Binary images derived from the annotations



Data Processing & Augmentation



Common Preprocessing

- **Resizing:** 2048x1024 → 256x256 px
- **Normalization:** Pixel intensity $[-1, 1]$

GAN Strategy

- Random Brightness $[0.8, 1.2]$
- Random Contrast $[0.8, 1.2]$
- Small Rotation $[+/- 10^\circ]$
- NO Geometric Distortions
(Preserves pixel-alignment)

YOLO Strategy

Geometric & Strong Augs:

- Mosaic (4-image grid)
- MixUp
- Random Flip

Generative Architecture (Pix2Pix)

Pix2Pix is a conditional GAN framework designed for general-purpose image-to-image translation. Unlike standard GANs that generate images from random noise, Pix2Pix generates images conditioned on a specific structure

Framework & Architecture

- Framework: PyTorch

Custom GeneratorUNet & DiscriminatorPatchGAN

- Generator: U-Net

Uses Skip Connections to preserve tooth contours

- Discriminator: PatchGAN

Penalizes structure at local NxN patch scale



Optimization

- Optimizer: Adam Solver

lr = 0,0002, beta1 = 0.5, beta2 = 0.999

- Loss Function Balance

$$\mathcal{L}_{Total} = \mathcal{L}_{cGAN} + 100 \cdot \mathcal{L}_{L1}$$

A high lamda (100) forces the Generator to prioritize correct anatomy over artistic details

- Epochs: 200

Training Pix2Pix

Epoch 50

Real X-Ray



Synthetic X-Ray

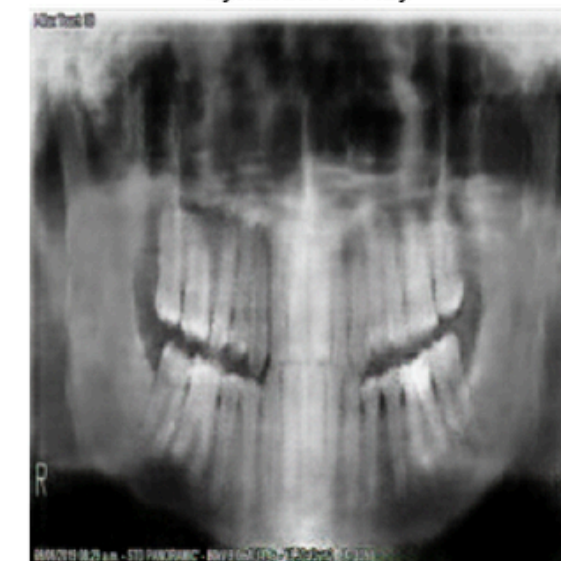


Epoch 200

Real X-Ray



Synthetic X-Ray



Epoch 100

Real X-Ray



Synthetic X-Ray



YOLO Segmentation

For the segmentation task, we chose YOLOv8. Unlike traditional detectors, it processes the entire image in a single pass to predict both Bounding Boxes and Pixel-wise Masks simultaneously. We selected the Nano-Segmentation (v8n-seg) variant for its balance between computational efficiency and high precision.

Training config. :

-Input size: 256x256px

-Classes: 32

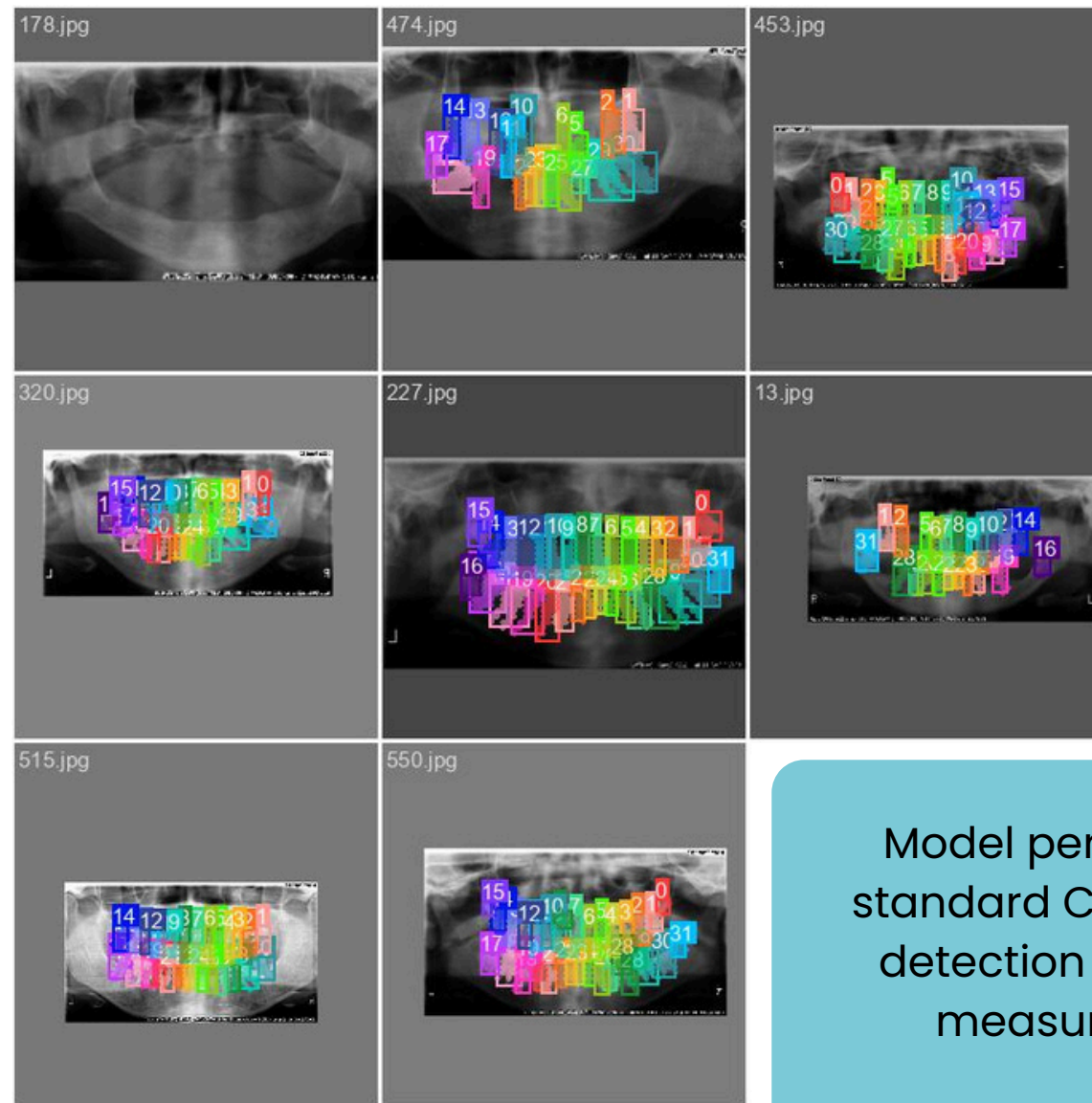
-Epochs: 100

-Batch Size: 16

-Optimizer: AdamW

-Momentum : 0.937

-Weight Decay: 0.0005 (prevent overfitting)



Post-processing to enhance X-Ray realism

- CLAHE (Contrast Limited Adaptive Histogram Equalization) → Enhances local contrast without amplifying background noise
- Denoising (Median Blur) → Removes the “salt-and-pepper” grain typical of GANs

Model performance was evaluated using standard COCO metrics: mAP@50 to assess detection robustness and mAP@50-95 to measure high-fidelity segmentation alignment.

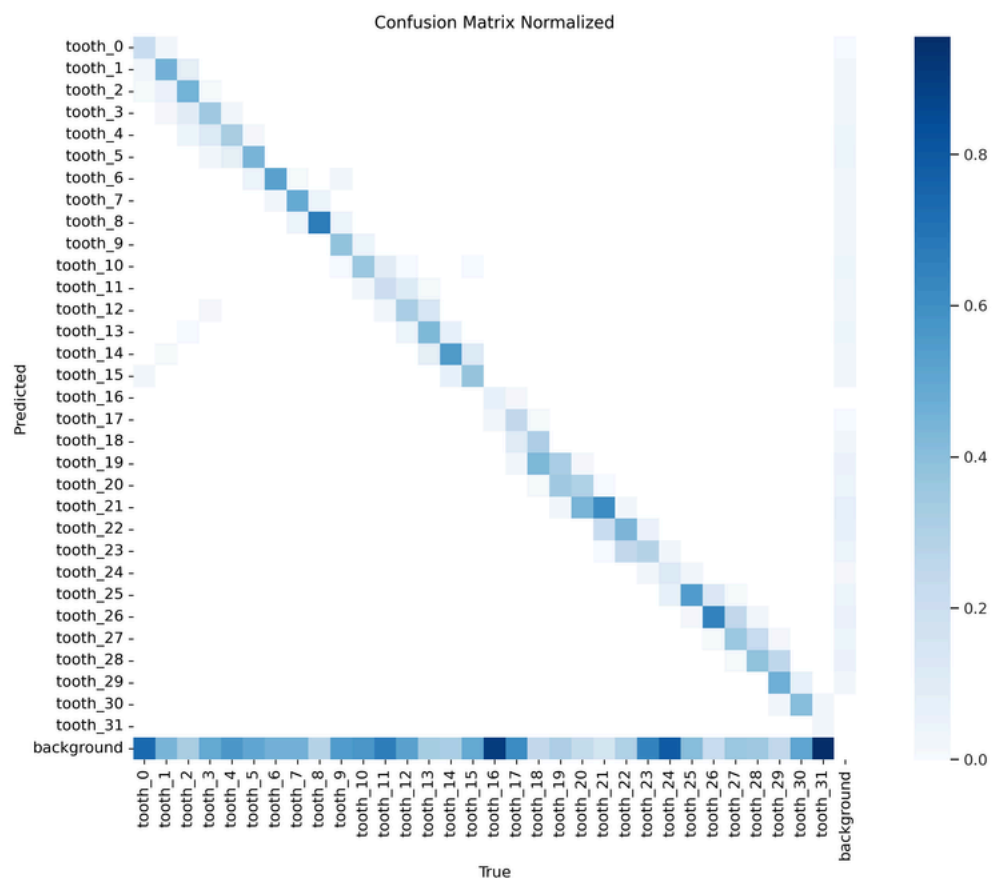
YOLO-based Quality Evaluation

Training Setup: we train the YOLO model exclusively on the Real Dataset (Original X-Rays)

Evaluation Step: We then test this model directly on Real data and the Synthetic Dataset (Generated Images)

Real Test Set: to establish a performance benchmark

Synthetic Test Set: To measure how well the model generalizes to generated images

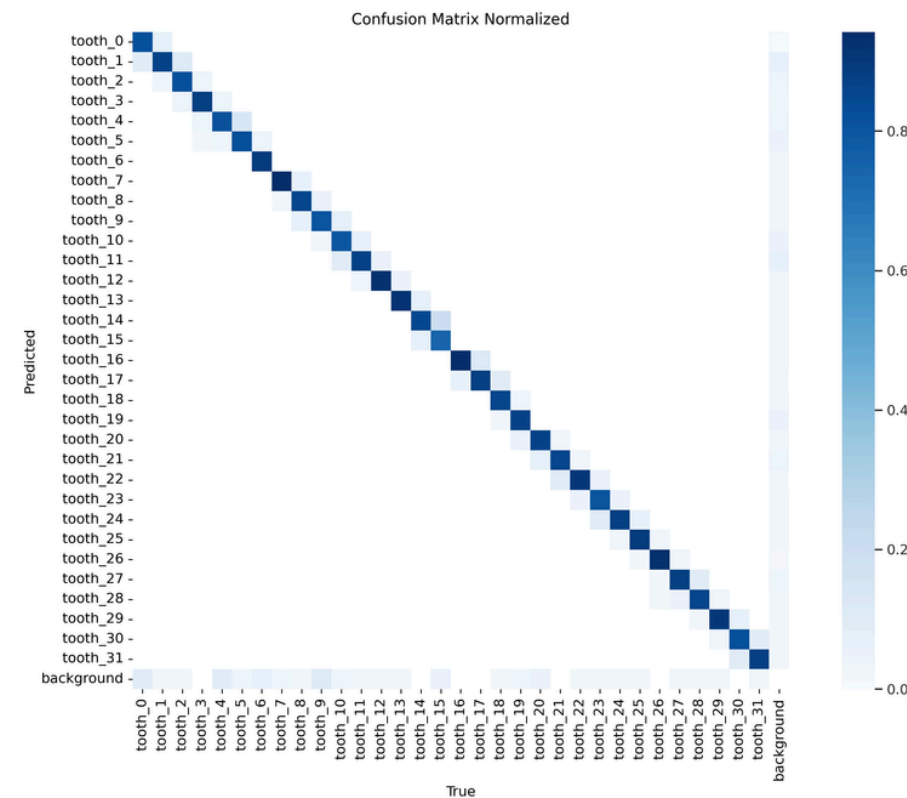
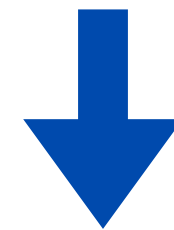


Data	mAP@50	mAP@50-95
Real	0.9011	0.5497
Synthetic	0.5636	0.2668

YOLO-based Quality Evaluation

Training Setup: we train the YOLO model on the Mixed Dataset (Original X-Rays + Generated Images)

Evaluation Step: We then test this model directly on Real data



Data	mAP@50	mAP@50-95
Real	0.9321	0.5854

+ 3.4% compared to the baseline

The inclusion of synthetic data improved the mAP@50. This confirms that the generated images act as an effective Data Augmentation technique, introducing valuable variability that helps the model generalize better and reduces overfitting on the limited real dataset.

“Weak” Generator

We replaced the standard U-Net generator with a basic Encoder-Decoder

The key difference is that in this way we remove Skip Connections:

- Pix2Pix (U-Net): Passes low-level features directly across layers to preserve edges
- Weak Variant: Forces all information through the bottleneck layer



- 32.5% compared to the synthetic baseline

Data	mAP@50	mAP@50-95
Weak	0.3800	0.1290



Future improvements



Generated images enhancement

While our current pipeline successfully proves the utility of synthetic data, it was constrained by available computational power. If we had access to larger GPU clusters, the immediate next step would be high-resolution synthesis (train the generator natively at 1024 x 1024 px).
Generating images natively at 1024x1024 px preserving the micro-anatomy of the teeth.

Models scaling

Upgrade from the lightweight YOLOv8-Nano to high-capacity architectures (e.g. YOLOv8-Large) to extract deeper features and maximize the mAP score

Upgrade the generator to more advanced architectures, such as a conditional Diffusion Model

References

- [1] [<https://www.kaggle.com/datasets/humansintheloop/teeth-segmentation-on-dental-x-ray-images/data>]
- [2] [<https://www.geeksforgeeks.org/deep-learning/generative-adversarial-network-gan/>]
- [3] [<https://medium.com/@ms.maryamrezaee/pix2pix-pytorch-implementation-what-is-it-and-how-to-do-it-f53bce51c84e>]
- [4] [<https://www.geeksforgeeks.org/python/clahe-histogram-equalization-opencv/>]
- [5] [<https://www.nature.com/articles/s41598-023-45290-1>]
- [6] [https://ieeexplore.ieee.org/abstract/document/10533619?casa_token=9nh2LirifacAAAAA:kjlr4vtSRapNdJqn6AnnHlmwLsmIPn-Ik9aLVIGkCAq3lZrPh9WFWN4ljA5mmfk_IS2lKje3pw]
- [7] [<https://arxiv.org/abs/2102.04306>]